Math 110 — Week 9

Trace and inner products (Notes Ch. 16–17, LADR 8D, 6A, HW9)

§1 Facts used from earlier weeks

V is finite-dimensional over $\mathbb{F}$, and $n = \dim V$.

(T1) **Change of basis.** If $A = \mathcal{M}_{\mathcal{B}}(T)$ and $A' = \mathcal{M}_{\mathcal{C}}(T)$ are the matrices of $T \in \mathcal{L}(V)$ in two bases, then $A' = P^{-1}AP$, where column j of the invertible matrix P holds the $\mathcal{B}$-coordinates of the j-th vector of $\mathcal{C}$.

Why. Both sides send the $\mathcal{C}$-coordinates of v to the $\mathcal{C}$-coordinates of Tv: P converts $\mathcal{C}$-coordinates to $\mathcal{B}$-coordinates, A applies T, and P^{-1} converts back.

(T2) **Jordan form.** Over $\mathbb{C}$, every $T \in \mathcal{L}(V)$ has a basis in which its matrix is upper-triangular, with the eigenvalues on the diagonal, each repeated as often as its algebraic multiplicity $\dim G(\lambda, T)$ (Week 8).

(T3) **Characteristic polynomial.** Over $\mathbb{C}$, $\chi_T(z) = \prod_{j=1}^{n}(z - \lambda_j)$, where $\lambda_1, \dots, \lambda_n$ are the eigenvalues listed with algebraic multiplicity (Week 8).

(T4) **Complex numbers.** $\overline{z+w} = \bar{z} + \bar{w}$, $\overline{zw} = \bar{z}\,\bar{w}$, $z\bar{z} = |z|^2$, $z + \bar{z} = 2\operatorname{Re} z$, and $\operatorname{Re} z \le |z|$. If a polynomial has real coefficients, its non-real roots come in conjugate pairs $\lambda, \bar{\lambda}$ with the same multiplicity.

Why. The first four are direct computations with $z = a + bi$. $\operatorname{Re} z = a \le \sqrt{a^2 + b^2} = |z|$. If p has real coefficients, then $\overline{p(\lambda)} = p(\bar{\lambda})$, so $p(\lambda) = 0 \iff p(\bar{\lambda}) = 0$; applying this to the quotient $p/(x - \lambda)(x - \bar{\lambda})$, which again has real coefficients, gives equal multiplicities.

(T5) **Projectors split V** (Week 6, HW6 P1). If $P^2 = P$, then $V = \operatorname{null} P \oplus \operatorname{range} P$, and $Pw = w$ for $w \in \operatorname{range} P$.

Why. $v = (v - Pv) + Pv$ with $P(v - Pv) = 0$; if $w = Pu$ is also in $\operatorname{null} P$, then $w = Pu = P^2u = Pw = 0$; and for $w = Pu$, $Pw = P^2u = Pu = w$.

(T6) **Minimal polynomial and Jordan blocks** (Week 8). Over $\mathbb{C}$, the exponent of $x - \lambda$ in m_T is the size of the largest Jordan block for λ; $J_k(\lambda)$ denotes the $k \times k$ Jordan block.

(T7) **A nonnegative continuous function with integral 0 is 0.** If $g\colon [a, b] \to \mathbb{R}$ is continuous, $g \ge 0$ and $\int_a^b g = 0$, then $g = 0$.

Why. If $g(c) > 0$, continuity gives an interval around c on which $g > g(c)/2$, and the integral over that interval alone is positive.

§2 This week

§2.1 The trace of a matrix (Ch. 16)

The question. Jordan form showed that a good basis reveals a lot about an operator. Is there a number we can read off *any* matrix of T that comes out the same in every basis?

Definition (Def. 41 / LADR 8.47)**.** The **trace** of an $n \times n$ matrix $A = (a_{ij})$ is the sum of its diagonal entries: $\operatorname{tr} A = a_{11} + \cdots + a_{nn}$.

For example,

$$\operatorname{tr}\begin{pmatrix}1&2&3\\4&5&6\\7&8&9\end{pmatrix} = 1+5+9 = 15.$$

Trace is linear: $\operatorname{tr}(A+B) = \operatorname{tr}A + \operatorname{tr}B$ and $\operatorname{tr}(cA) = c\operatorname{tr}A$, because the diagonal of $A+B$ is the sum of the diagonals. It is not multiplicative: $\operatorname{tr}(AB)$ is usually not $\operatorname{tr}A \cdot \operatorname{tr}B$ (take $A = B = I_2$: $\operatorname{tr}(AB) = 2$, $\operatorname{tr}A \cdot \operatorname{tr}B = 4$). What it does satisfy is the following.

Theorem (Prop. 78 / LADR 8.49: the trace of a product does not depend on the order)
If A is $m \times n$ and B is $n \times m$, then $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

The idea

Write out both traces as sums of products of entries. The diagonal entry $(AB)_{ii}$ is row i of A times column i of B, which is $\sum_j a_{ij}b_{ji}$. So $\operatorname{tr}(AB)$ is the sum of all the products $a_{ij}b_{ji}$. The same products appear in $\operatorname{tr}(BA)$, only grouped by j instead of by i. Nothing else is needed: the two traces are one finite sum added in two different orders.

Proof. 1. By the definition of matrix multiplication, $(AB)_{ii} = \sum_{j=1}^{n} a_{ij}b_{ji}$ for $i = 1, \dots, m$, so

$$\operatorname{tr}(AB) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}b_{ji}.$$

2. In the same way $(BA)_{jj} = \sum_{i=1}^{m} b_{ji}a_{ij}$ for $j = 1, \dots, n$, so

$$\operatorname{tr}(BA) = \sum_{j=1}^{n}\sum_{i=1}^{m} b_{ji}a_{ij}.$$

3. Both are the sum of the mn numbers $a_{ij}b_{ji}$, over all pairs (i, j). A finite sum does not depend on the order of its terms, so the two traces are equal.

□

	$j=1$	$j=2$	$j=3$	add along the row
$i=1$	$a_{11}b_{11}$	$a_{12}b_{21}$	$a_{13}b_{31}$	$(AB)_{11}$
$i=2$	$a_{21}b_{12}$	$a_{22}b_{22}$	$a_{23}b_{32}$	$(AB)_{22}$
add down the column	$(BA)_{11}$	$(BA)_{22}$	$(BA)_{33}$	total of all six

The case $m = 2$, $n = 3$. The six products $a_{ij}b_{ji}$, one for each pair (i, j). Adding along the rows gives the diagonal entries of AB; adding down the columns gives the diagonal entries of BA. Both totals are the sum of all six products.

Example: the trace of a product computed both ways. For

$$A = \begin{pmatrix}1&2&0\\0&1&3\end{pmatrix}, \quad B = \begin{pmatrix}1&0\\2&1\\0&1\end{pmatrix}: \qquad AB = \begin{pmatrix}5&2\\2&4\end{pmatrix}, \quad BA = \begin{pmatrix}1&2&0\\2&5&3\\0&1&3\end{pmatrix},$$

and both traces are 9, although AB is 2×2 and BA is 3×3.

Similar matrices have the same trace. If P is invertible, apply Prop. 78 to the two matrices P^{-1} and AP:

$$\operatorname{tr}(P^{-1}AP) = \operatorname{tr}\big(P^{-1}(AP)\big) = \operatorname{tr}\big((AP)P^{-1}\big) = \operatorname{tr}A.$$

Cyclic order only. Applying Prop. 78 to A and BC gives $\operatorname{tr}(ABC) = \operatorname{tr}(BCA)$, and once more $= \operatorname{tr}(CAB)$: moving the first factor to the end is allowed. Swapping two factors is not (Traps below). The **matrix unit** E_{ij} has a single 1 in position (i, j) and 0 elsewhere.

The question answered. Yes: tr of the matrix of T is the same in every basis, because the matrices are similar and similar matrices have the same trace.

Shortcuts

- $\operatorname{tr}(AB)$ **without multiplying**: $\operatorname{tr}(AB) = \sum_{i,j} a_{ij} b_{ji}$. Reason: Prop. 78, step 1. Condition: A is $m \times n$, B is $n \times m$. Instance: for the A, B above, $1 \cdot 1 + 2 \cdot 2 + 0 + 0 + 1 \cdot 1 + 3 \cdot 1 = 9$.
- **Sum of squares of entries**: $\operatorname{tr}(A^{\mathsf{T}} A) = \sum_{i,j} a_{ij}^2$. Reason: the previous item applied to A^{T} and A gives $\sum_{i,j} (A^{\mathsf{T}})_{ij} a_{ji} = \sum_{i,j} a_{ji}^2$. Instance: $A = \left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$ gives $1 + 4 + 9 + 16 = 30$.
- **Cyclic moves**: $\operatorname{tr}(ABC) = \operatorname{tr}(BCA) = \operatorname{tr}(CAB)$ and $\operatorname{tr}(P^{-1} A P) = \operatorname{tr} A$. Reason: Prop. 78 applied to A and BC. Instance: every matrix similar to $\left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$ has trace 5.
- **Reading off an entry with a trace**: $\operatorname{tr}(E_{ji} A) = a_{ij}$, so if $\operatorname{tr}(SA) = 0$ for every S, then $A = 0$. Reason: $E_{ji} A$ has row i of A in row j and zeros elsewhere, so its only diagonal entry is a_{ij}. Instance: for $A = \left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$, $\operatorname{tr}(E_{21} A) = a_{12} = 2$.

Traps

- $\operatorname{tr}(AB) \neq \operatorname{tr} A \cdot \operatorname{tr} B$ in general: $A = B = I_2$ gives 2 and 4.
- Only cyclic moves are allowed:

$$\operatorname{tr}(E_{11} E_{12} E_{21}) = \operatorname{tr} E_{11} = 1, \qquad \operatorname{tr}(E_{11} E_{21} E_{12}) = \operatorname{tr}(0 \cdot E_{12}) = 0.$$

- Equal traces do not make matrices similar: $\left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ and 0 both have trace 0, but the only matrix similar to 0 is 0.

§2.2 The trace of an operator (Ch. 16)

The question. $T(x, y) = (x + 2y,\ 3x + 4y)$ has diagonal $1, 4$ in the standard basis, and in the basis $(1, 1), (1, -1)$ its matrix is $\left[\begin{smallmatrix} 5 & -1 \\ -2 & 0 \end{smallmatrix}\right]$, with diagonal $5, 0$. Both diagonals add up to 5. Is that true in every basis, and can the number 5 be found from the eigenvalues $\frac{5 \pm \sqrt{33}}{2}$ alone?

Definition (Def. 42 / LADR 8.50, 8.51)**.** For $T \in \mathcal{L}(V)$, choose any basis $\mathcal{B}$ of V and define $\operatorname{tr} T = \operatorname{tr} \mathcal{M}_{\mathcal{B}}(T)$.

This does not depend on the basis: by (T1) the matrices of T in two bases are similar, and similar matrices have the same trace. So $\operatorname{tr} T$ is computed from coordinates but is a property of T alone. In particular $\operatorname{tr} I_V = n$, and tr is linear on $\mathcal{L}(V)$: in a fixed basis $\mathcal{M}(S + T) = \mathcal{M}(S) + \mathcal{M}(T)$ and $\mathcal{M}(cT) = c\mathcal{M}(T)$, and the trace of matrices is linear.

Theorem (Prop. 79 / LADR 8.52: over $\mathbb{C}$, trace is the sum of the eigenvalues)

If V is a complex vector space and $T \in \mathcal{L}(V)$, then $\operatorname{tr} T$ is the sum of the eigenvalues of T, each counted with its algebraic multiplicity. If the distinct eigenvalues are $\lambda_1, \ldots, \lambda_r$ with algebraic multiplicities $m_1, \ldots, m_r$, then $\operatorname{tr} T = m_1 \lambda_1 + \cdots + m_r \lambda_r$.

The idea

The trace may be computed in any basis, so compute it in the basis where the eigenvalues are visible: the Jordan basis, whose diagonal is exactly the list of eigenvalues with multiplicities.

Proof. 1. By (T2) there is a basis in which $\mathcal{M}(T)$ is upper-triangular with diagonal entries $\lambda_1, \ldots, \lambda_1, \ldots, \lambda_r, \ldots, \lambda_r$, where λ_k appears m_k times.

2. $\operatorname{tr} T$ is the trace of this matrix, which is the sum of its diagonal entries, $m_1 \lambda_1 + \cdots + m_r \lambda_r$.

□

Example 64. The eigenvalues of

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix}$$

are roots of the cubic $\lambda^3 - 12\lambda^2 + 15\lambda - 2$, which are unpleasant to write down. Their sum is still immediate:

$$\lambda_1 + \lambda_2 + \lambda_3 = \operatorname{tr} A = 1 + 2 + 9 = 12.$$

- **Trace in the characteristic polynomial.** Expanding $\chi_T(z) = \prod_{j=1}^{n}(z - \lambda_j)$ (T3), the terms with z^{n-1} come from choosing z in all factors but one and $-\lambda_j$ in the remaining one. So the coefficient of z^{n-1} is $-(\lambda_1 + \cdots + \lambda_n) = -\operatorname{tr} T$. In the cubic of Ex. 64 this coefficient is -12.
- **Real matrices.** A real matrix A is also an operator on $\mathbb{C}^n$, and its trace is the same number. Its non-real eigenvalues come in pairs $\lambda, \bar{\lambda}$ with equal algebraic multiplicity. Reason: for $v \in \mathbb{C}^n$ let $\bar{v}$ be the vector of conjugated entries. Since A is real, $\overline{(A - \lambda I)^k v} = (A - \bar{\lambda} I)^k \bar{v}$, so $v \in G(\lambda, A)$ if and only if $\bar{v} \in G(\bar{\lambda}, A)$. Conjugation carries a basis $v_1, \ldots, v_d$ of $G(\lambda, A)$ to an independent list in $G(\bar{\lambda}, A)$ (conjugate any relation $\sum c_j \bar{v}_j = 0$ to get $\sum \bar{c}_j v_j = 0$), so $\dim G(\bar{\lambda}, A) \geq \dim G(\lambda, A)$, and with λ and $\bar{\lambda}$ exchanged the two are equal. Each pair contributes $\lambda + \bar{\lambda} = 2 \operatorname{Re} \lambda$ to the trace. Example: rotation of $\mathbb{R}^2$ by θ has eigenvalues $e^{\pm i\theta}$ and trace $2\cos\theta$, which is the sum of its diagonal entries $\cos\theta + \cos\theta$.

The question answered. Yes, the sum of the diagonal entries is $\operatorname{tr} T$ in every basis (Def. 42), and it is the sum of the eigenvalues (Prop. 79):

$$\frac{5 + \sqrt{33}}{2} + \frac{5 - \sqrt{33}}{2} = 5.$$

Shortcuts

- **Sum of the eigenvalues** (over $\mathbb{C}$, with algebraic multiplicity) $= \operatorname{tr} T$. Reason: Prop. 79. Instance: Ex. 64, sum 12; a nilpotent T has trace 0.
- 2×2 **characteristic polynomial**: for $A = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$, $\chi_A(z) = z^2 - (a+d)z + (ad - bc)$, so $\lambda_1 + \lambda_2 = a + d$ and $\lambda_1 \lambda_2 = ad - bc$. Reason: Week 8 (the 2×2 case of Cayley–Hamilton). Instance: $\left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$ has $\chi(z) = z^2 - 5z - 2$ and eigenvalues $\frac{5 \pm \sqrt{33}}{2}$.
- **Traces of powers**: over $\mathbb{C}$, $\operatorname{tr}(T^k) = \lambda_1^k + \cdots + \lambda_n^k$. Reason: in the basis of (T2), $\mathcal{M}(T)^k$ is upper-triangular with diagonal λ_j^k. Instance: for $A = \left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$, $A^2 = \left[\begin{smallmatrix} 7 & 10 \\ 15 & 22 \end{smallmatrix}\right]$ has trace $29 = (\lambda_1 + \lambda_2)^2 - 2\lambda_1\lambda_2 = 25 + 4$.
- **Trace from** χ_T: $-\operatorname{tr} T$ is the coefficient of z^{n-1}. Reason: the bullet above. Instance: Ex. 64, coefficient -12.
- **Trace of a projector** ($P^2 = P$): $\operatorname{tr} P = \dim \operatorname{range} P$. Reason: in a basis of $\operatorname{null} P$ followed by a basis of $\operatorname{range} P$ (T5), $\mathcal{M}(P)$ is diagonal with entries 0 and 1. Instance: $P(x, y) = (x + y, 0)$ has trace 1 and range the x-axis.
- **Real matrices**: a non-real eigenvalue λ comes with $\bar{\lambda}$, together contributing $2 \operatorname{Re} \lambda$. Reason: the bullet above. Instance: a real 3×3 matrix with eigenvalues 3 and $2 + i$ has trace $3 + 4 = 7$.

Traps

- Trace 0 does not mean nilpotent: $\operatorname{diag}(1, -1)$.
- Over $\mathbb{R}$ the trace is not the sum of the real eigenvalues: $\left[\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right]$ has no real eigenvalue but trace 2 (its complex eigenvalues are $1 \pm i$).
- Count algebraic, not geometric, multiplicity: $\left[\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right]$ has a single line of eigenvectors, yet its trace is $4 = 2 + 2$.

§2.3 Commutators have trace zero (Ch. 16)

The question. For the 2×2 matrix units, $E_{12}E_{21} - E_{21}E_{12} = \left[\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right]$. Can $AB - BA$ equal the identity I_2 for some 2×2 matrices A, B?

Theorem (Prop. 80: commutators have trace 0)
For $S, T \in \mathcal{L}(V)$, $\operatorname{tr}(ST - TS) = 0$. Consequently, if $V \neq \{0\}$ and $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$, there are no $S, T \in \mathcal{L}(V)$ with $ST - TS = I$.

The idea

ST and TS are different operators, but Prop. 78 says they have the same trace, so their difference has trace 0. The identity has trace $n \neq 0$, so it cannot be a difference of this kind.

Proof. 1. Trace is linear, so $\operatorname{tr}(ST - TS) = \operatorname{tr}(ST) - \operatorname{tr}(TS)$. In any basis, $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$ and $\mathcal{M}(TS) = \mathcal{M}(T)\mathcal{M}(S)$, and these have the same trace by Prop. 78. So $\operatorname{tr}(ST - TS) = 0$.

2. $\operatorname{tr} I = n$, which is not 0 in $\mathbb{R}$ or $\mathbb{C}$ because $n \geq 1$. So $ST - TS = I$ would give $0 = n$, which is false.

□

Finite dimension is needed. On the infinite-dimensional space $\mathcal{P}(\mathbb{R})$, let D be differentiation and M multiplication by x. The product rule gives $D(xp) = p + xp'$, so $(DM - MD)p = p$ for every p: here $DM - MD = I$. There is no trace on $\mathcal{P}(\mathbb{R})$ to rule it out.

The question answered. No: $\operatorname{tr}(AB - BA) = 0$ but $\operatorname{tr} I_2 = 2$ (Prop. 80).

Shortcuts

- **Impossible commutator**: $ST - TS = X$ has no solution if $\operatorname{tr} X \neq 0$. Reason: Prop. 80. Condition: finite dimension. Instance: no 2×2 A, B have $AB - BA = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$ (trace 2).

Traps

- $\operatorname{tr}(ST - TS) = 0$ does not mean $ST = TS$: $E_{12}E_{21} - E_{21}E_{12} = \operatorname{diag}(1, -1) \neq 0$.
- $ST - TS = I$ is impossible only in finite dimension: $DM - MD = I$ on $\mathcal{P}(\mathbb{R})$.

§2.4 Inner products (Ch. 17)

The question. So far the course has been algebraic: nothing has had a length or an angle. Lengths and angles in $\mathbb{R}^2$ come from the dot product. Which properties of the dot product are needed to do geometry in an arbitrary vector space, including a complex one?

From now on $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$. Over $\mathbb{R}$, ignore every complex conjugation below.

Definition (Def. 43 / LADR 6.2)**.** An **inner product** on V is a function $\langle \cdot, \cdot \rangle \colon V \times V \to \mathbb{F}$ such that for all $u, v, w \in V$ and $\lambda \in \mathbb{F}$:

(i) *positive definite:* $\langle v, v \rangle > 0$ for every $v \neq 0$;

(ii) *linear in the first slot:* $\langle \lambda u + v, w \rangle = \lambda \langle u, w \rangle + \langle v, w \rangle$;

(iii) *conjugate symmetric:* $\langle u, v \rangle = \overline{\langle v, u \rangle}$.

An **inner product space** is a vector space with a chosen inner product.

Why the conjugate. On $\mathbb{R}^n$ the dot product $\sum x_j y_j$ is positive on nonzero vectors, because $\sum x_j^2 > 0$. Copied to $\mathbb{C}^n$ without change, it fails: $(i,1)\cdot(i,1) = i^2 + 1^2 = 0$ although $(i,1) \neq 0$. The fix is to conjugate the second vector, because $z\bar{z} = |z|^2 \geq 0$ (T4).

Consequences of the axioms.

- $\langle v, v\rangle$ is real, by (iii) with $u = v$.
- *Conjugate-linear in the second slot:* $\langle u, \lambda v + w\rangle = \bar{\lambda}\langle u, v\rangle + \langle u, w\rangle$. Proof: by (iii), (ii), and (iii) again with (T4),

$$\langle u, \lambda v + w\rangle = \overline{\langle \lambda v + w, u\rangle} = \overline{\lambda\langle v, u\rangle + \langle w, u\rangle} = \bar{\lambda}\,\overline{\langle v, u\rangle} + \overline{\langle w, u\rangle} = \bar{\lambda}\langle u, v\rangle + \langle u, w\rangle.$$

- $\langle 0, v\rangle = \langle v, 0\rangle = 0$: by (ii), $\langle 0, v\rangle = \langle 0 \cdot 0, v\rangle = 0\langle 0, v\rangle = 0$, and then (iii) gives the other.

Examples 65, 66 and two forms on $\mathbb{R}^2$.

- Ex. 65, the **dot product** on $\mathbb{F}^n$: $\langle w, z\rangle = w_1\bar{z}_1 + \cdots + w_n\bar{z}_n$. Positive definite because $\langle w, w\rangle = |w_1|^2 + \cdots + |w_n|^2$, which is 0 only when every $w_j = 0$. Properties (ii) and (iii) follow from (T4) term by term.
- Ex. 66, the **L^2 product** on continuous functions $[0,1] \to \mathbb{C}$: $\langle f, g\rangle = \int_0^1 f(x)\overline{g(x)}\,dx$. Linearity in the first slot comes from linearity of the integral; conjugate symmetry from the rule that the conjugate of an integral is the integral of the conjugate, applied to $h = f\bar{g}$. Positive definite: $\langle f, f\rangle = \int_0^1 |f|^2$, and if this is 0 then $|f|^2 = 0$ by (T7), so $f = 0$. The same works on any interval $[a, b]$; on an unbounded interval the integrals must be required to converge.
- Weights: on $\mathbb{R}^2$, $\langle x, y\rangle = x_1y_1 + x_1y_2 + x_2y_1 + 2x_2y_2$ is an inner product. Linearity and symmetry are clear; for positivity complete the square:

$$\langle x, x\rangle = x_1^2 + 2x_1x_2 + 2x_2^2 = (x_1 + x_2)^2 + x_2^2,$$

  which is 0 only if $x_2 = 0$ and $x_1 = 0$. More generally, $c_1w_1\bar{z}_1 + \cdots + c_nw_n\bar{z}_n$ with every $c_j > 0$ is an inner product on $\mathbb{F}^n$ (positivity: $\sum c_j|w_j|^2 > 0$ unless all $w_j = 0$); if some $c_j \leq 0$, the vector e_j gives $\langle e_j, e_j\rangle = c_j \leq 0$.
- A non-example: $\langle x, y\rangle = x_1y_2 + x_2y_1$ on $\mathbb{R}^2$ is linear and symmetric, but $\langle (1,-1), (1,-1)\rangle = -2 < 0$.

The question answered. Positivity, linearity in the first slot and conjugate symmetry (Def. 43); the conjugation is what keeps $\langle v, v\rangle > 0$ over $\mathbb{C}$.

Shortcuts

- 2×2 **positivity test**: $\langle x, y\rangle = ax_1y_1 + b(x_1y_2 + x_2y_1) + cx_2y_2$ is an inner product on $\mathbb{R}^2$ if and only if $a > 0$ and $ac - b^2 > 0$. Reason: for $a > 0$,

$$\langle x, x\rangle = a\Big(x_1 + \frac{b}{a}x_2\Big)^2 + \frac{ac - b^2}{a}\,x_2^2;$$

  if $a \leq 0$ then $x = e_1$ gives $\langle x, x\rangle = a \leq 0$; if $ac - b^2 \leq 0 < a$ then $x = (-b, a)$ gives $\langle x, x\rangle = a(ac - b^2) \leq 0$. Instance: the weighted example has $a = 1$, $b = 1$, $c = 2$, and $ac - b^2 = 1 > 0$.
- **Find a bad vector**: to disprove positivity, try a vector that makes a cross term negative. Instance: $(1, -1)$ for $x_1y_2 + x_2y_1$.
- **Scalars out of the second slot are conjugated**: $\langle u, \lambda v\rangle = \bar{\lambda}\langle u, v\rangle$. Reason: conjugate-linearity above. Instance: in $\mathbb{C}$ with $\langle z, w\rangle = z\bar{w}$, $\langle 1, i\rangle = -i$.

Traps

- Linear and symmetric is not enough: $x_1y_2 + x_2y_1$ gives $\langle (1,-1), (1,-1)\rangle = -2$.
- Without conjugation, the dot product on $\mathbb{C}^n$ is not positive: $(i,1)\cdot(i,1) = 0$.
- $\langle u, \lambda v\rangle = \lambda\langle u, v\rangle$ is false over $\mathbb{C}$: $\langle 1, i\cdot 1\rangle = -i$, not i.

§2.5 The norm and the Cauchy–Schwarz inequality (Ch. 17)

The question. For $u = (1, 2)$, $v = (3, 1)$ in $\mathbb{R}^2$, $|u \cdot v| = 5$ while $\|u\|\,\|v\| = \sqrt{5} \cdot \sqrt{10} \approx 7.07$. Is $|\langle u, v\rangle| \le \|u\|\,\|v\|$ in every inner product space (for example for $\int_0^1 f\bar{g}$), and when is it an equality?

Definition (Def. 44)**.** A **norm** on V is a function $\|\cdot\| : V \to \mathbb{R}$ with (i) $\|v\| > 0$ for $v \neq 0$; (ii) $\|\lambda v\| = |\lambda|\,\|v\|$; (iii) the **triangle inequality** $\|u + v\| \le \|u\| + \|v\|$.

An inner product gives the candidate $\|v\| = \sqrt{\langle v, v\rangle}$, the length. (For the dot product on $\mathbb{R}^2$ this is $\sqrt{x_1^2 + x_2^2}$.) Three facts about it are needed first. Scaling works as expected: by linearity in the first slot and conjugate-linearity in the second, $\|\lambda v\|^2 = \langle \lambda v, \lambda v\rangle = \lambda\bar{\lambda}\langle v, v\rangle = |\lambda|^2\|v\|^2$, so $\|\lambda v\| = |\lambda|\,\|v\|$. Vectors u, v are **orthogonal** if $\langle u, v\rangle = 0$. Then 0 is orthogonal to every vector, because $\langle 0, v\rangle = 0$; and 0 is the only vector orthogonal to itself, because $\langle v, v\rangle = 0$ forces $v = 0$ by positivity.

Pythagoras. If $\langle u, v\rangle = 0$, then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$. Proof: expand by linearity in each slot,

$$\|u + v\|^2 = \langle u + v, u + v\rangle = \langle u, u\rangle + \langle u, v\rangle + \langle v, u\rangle + \langle v, v\rangle = \|u\|^2 + 0 + 0 + \|v\|^2.$$

Splitting off the part along v. If $v \neq 0$ and $c = \langle u, v\rangle / \|v\|^2$, then $w = u - cv$ is orthogonal to v. Proof:

$$\langle w, v\rangle = \langle u, v\rangle - c\langle v, v\rangle = \langle u, v\rangle - \langle u, v\rangle = 0.$$

So $u = cv + w$ is u split into a piece along v and a piece perpendicular to v.

Theorem (Prop. 82 / LADR 6.14: Cauchy–Schwarz inequality)
For all $u, v \in V$, $|\langle u, v\rangle| \le \|u\|\,\|v\|$, with equality if and only if one of u, v is a scalar multiple of the other.

The idea

In $\mathbb{R}^2$, drop the perpendicular from the tip of u to the line through v. The foot of the perpendicular is no farther from 0 than u is, because u is the hypotenuse of a right triangle. To prove this without angles, split u into its piece cv along v and its perpendicular piece w. By Pythagoras $\|u\|^2 = \|cv\|^2 + \|w\|^2 \ge \|cv\|^2$, and $\|cv\|$ turns out to be exactly $|\langle u, v\rangle|/\|v\|$. Equality means the perpendicular piece is 0.

Proof. 1. If $v = 0$, both sides are 0, and $v = 0 \cdot u$ is a multiple of u. From now on $v \neq 0$.

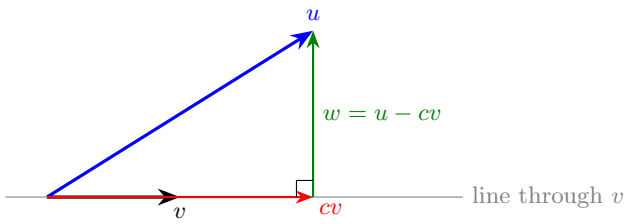
2. Let $c = \langle u, v\rangle / \|v\|^2$ and $w = u - cv$. Then $\langle w, v\rangle = 0$, so cv and w are orthogonal, and Pythagoras gives

$$\|u\|^2 = \|cv\|^2 + \|w\|^2 = |c|^2\|v\|^2 + \|w\|^2 \ge |c|^2\|v\|^2 = \frac{|\langle u, v\rangle|^2}{\|v\|^2}.$$

3. Multiply by $\|v\|^2$ and take square roots: $|\langle u, v\rangle| \le \|u\|\,\|v\|$.

4. Equality holds exactly when $\|w\|^2 = 0$, that is, when $u = cv$. Conversely, if $u = av$ for a scalar a, then $|\langle u, v\rangle| = |a|\,\|v\|^2 = \|u\|\,\|v\|$.

(The notes expand $\langle z, z\rangle \ge 0$ for $z = cv - u = -w$; it is the same computation.) □



$u = cv + w$ with $w \perp v$. The hypotenuse u is at least as long as the side cv, and $\|cv\| = |c|\,\|v\| = |\langle u, v\rangle|/\|v\|$. That is the Cauchy–Schwarz inequality. Equality means $w = 0$: u lies on the line through v.

Example 67: bounding an integral. The function $\sqrt{\sin x}$ has no elementary antiderivative; numerically its integral over $[0,\pi]$ is about 2.396. With the L^2 product on $[0,\pi]$, $f=\sqrt{\sin x}$ and $g=1$:

$$\int_0^\pi \sqrt{\sin x}\,dx = \langle f,g\rangle \le \|f\|\,\|g\| = \sqrt{\int_0^\pi \sin x\,dx}\cdot\sqrt{\int_0^\pi 1\,dx} = \sqrt{2}\cdot\sqrt{\pi} = \sqrt{2\pi} \approx 2.507.$$

Theorem (Prop. 81 / LADR 6.7, 6.17: an inner product gives a norm)
$\|v\| = \sqrt{\langle v,v\rangle}$ is a norm on V.

The idea

Positivity and scaling come straight from the axioms. For the triangle inequality, expand $\|u+v\|^2$: the cross terms add up to $2\,\mathrm{Re}\langle u,v\rangle$, and Cauchy–Schwarz bounds them by $2\|u\|\,\|v\|$, which is exactly the cross term of $(\|u\|+\|v\|)^2$.

Proof. 1. *Positive.* If $v\neq 0$ then $\langle v,v\rangle > 0$, so $\|v\| > 0$.

2. *Scaling.* $\|\lambda v\| = |\lambda|\,\|v\|$ was shown before Pythagoras.

3. *Triangle inequality.* Expand, using $\langle v,u\rangle = \overline{\langle u,v\rangle}$ and $z+\bar z = 2\,\mathrm{Re}\,z$ (T4):

$$\|u+v\|^2 = \|u\|^2 + \langle u,v\rangle + \langle v,u\rangle + \|v\|^2 = \|u\|^2 + 2\,\mathrm{Re}\langle u,v\rangle + \|v\|^2.$$

4. Since $\mathrm{Re}\,z \le |z|$ (T4) and by Cauchy–Schwarz,

$$\|u\|^2 + 2\,\mathrm{Re}\langle u,v\rangle + \|v\|^2 \le \|u\|^2 + 2\|u\|\,\|v\| + \|v\|^2 = \big(\|u\|+\|v\|\big)^2, \quad \text{so} \quad \|u+v\| \le \|u\| + \|v\|.$$

□

Equality in the triangle inequality needs equality in both steps of 4: u, v dependent and $\mathrm{Re}\langle u,v\rangle = |\langle u,v\rangle|$. For $v\neq 0$ that means $u = av$ with $a \ge 0$: if $u=av$, then $\langle u,v\rangle = a\|v\|^2$, and $\mathrm{Re}\langle u,v\rangle = |\langle u,v\rangle|$ means $\mathrm{Re}\,a = |a|$, that is, a is real and ≥ 0. The two vectors point the same way.

The question answered. Yes, in every inner product space (Prop. 82), with equality exactly when one vector is a multiple of the other.

Shortcuts

- **Cauchy–Schwarz with a chosen vector**: $(a_1b_1+\cdots+a_nb_n)^2 \le (a_1^2+\cdots+a_n^2)(b_1^2+\cdots+b_n^2)$. Reason: Prop. 82 for the dot product. Instances: with $b=(1,\dots,1)$,

  $$(a_1+\cdots+a_n)^2 \le n(a_1^2+\cdots+a_n^2);$$

  with $b=(1,2,3)$, $(a+2b+3c)^2 \le 14(a^2+b^2+c^2)$; for integrals, $\left(\int fg\right)^2 \le \int f^2 \int g^2$, often with $g=1$ (Ex. 67).
- **Equality in Cauchy–Schwarz**: exactly when one vector is a multiple of the other; name the multiple. Reason: Prop. 82. Instance: equality in $(a+b+c)^2 \le 3(a^2+b^2+c^2)$ only for $a=b=c$.
- **Expand a squared norm**: $\|u+v\|^2 = \|u\|^2 + 2\,\mathrm{Re}\langle u,v\rangle + \|v\|^2$. Reason: Prop. 81, step 3. Instance: for $u=(1,2)$, $v=(3,-1)$, $\|u+v\|^2 = 5+2+10 = 17$.

Traps

- Pythagoras needs orthogonality: $u=v=e_1$ gives $\|u+v\|^2 = 4 \neq 1+1$.
- Over $\mathbb{C}$, $\mathrm{Re}\langle u,v\rangle = 0$ already gives $\|u+v\|^2 = \|u\|^2+\|v\|^2$, but that is not orthogonality: in $\mathbb{C}$, $u=1$, $v=i$ give $|1+i|^2 = 2 = 1+1$ while $\langle u,v\rangle = -i \neq 0$.

- The triangle inequality is an equality only for a *nonnegative* multiple: $u = e_1$, $v = -e_1$ are dependent, but $\|u+v\| = 0 < 2$.

§2.6 Which norms come from inner products (Ch. 17)

The question. For the dot product and $u = (1,2)$, $v = (3,-1)$,

$$\|u+v\|^2 + \|u-v\|^2 = 17 + 13 = 30 = 2\cdot 5 + 2\cdot 10 = 2\|u\|^2 + 2\|v\|^2.$$

For the norm $\max(|x|,|y|)$ and $u = e_1$, $v = e_2$ the two sides are 2 and 4. Which norms come from an inner product, and how is the inner product recovered from the norm?

Theorem (Prop. 83 / LADR 6.21, 6A.28: parallelogram law and polarization)

A norm $\|\cdot\|$ on V comes from an inner product if and only if (the "if" direction is cited, not proved: the Jordan–von Neumann theorem) it satisfies the **parallelogram law**

$$\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2 \quad \text{for all } u, v.$$

In that case the inner product is recovered from the norm by the **polarization identity**

$$\langle u,v\rangle = \frac{\|u+v\|^2 - \|u-v\|^2}{4} \quad (\mathbb{F}=\mathbb{R}), \qquad \langle u,v\rangle = \frac{\|u+v\|^2 - \|u-v\|^2 + i\|u+iv\|^2 - i\|u-iv\|^2}{4} \quad (\mathbb{F}=\mathbb{C}).$$

The idea

Geometrically: the two diagonals of a parallelogram have squared lengths adding up to the squared lengths of the four sides. For a norm from an inner product, expanding $\|u \pm v\|^2$ gives the same cross term with opposite signs; adding cancels it (parallelogram law) and subtracting isolates it (polarization).

Proof of the parallelogram law and polarization for a norm from an inner product. 1. As in the proof of Prop. 81,

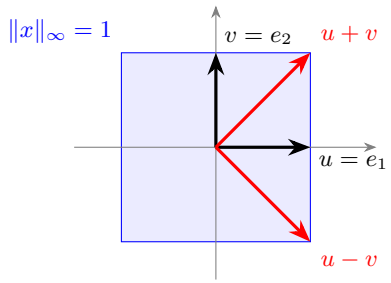
$$\|u+v\|^2 = \|u\|^2 + 2\,\mathrm{Re}\langle u,v\rangle + \|v\|^2, \qquad \|u-v\|^2 = \|u\|^2 - 2\,\mathrm{Re}\langle u,v\rangle + \|v\|^2.$$

2. Adding the two lines gives the parallelogram law.
3. Subtracting gives $\|u+v\|^2 - \|u-v\|^2 = 4\,\mathrm{Re}\langle u,v\rangle$. Over $\mathbb{R}$, $\langle u,v\rangle$ is real, which is the real polarization identity.
4. Over $\mathbb{C}$, apply step 3 to u and iv, using $\langle u, iv\rangle = \bar{i}\langle u,v\rangle = -i\langle u,v\rangle$ and $\mathrm{Re}(-iz) = \mathrm{Im}\, z$ (for $z = a+bi$, $-iz = b - ai$):
$$\|u+iv\|^2 - \|u-iv\|^2 = 4\,\mathrm{Re}\langle u, iv\rangle = 4\,\mathrm{Re}\big(-i\langle u,v\rangle\big) = 4\,\mathrm{Im}\langle u,v\rangle.$$
So the numerator of the complex formula is $4\,\mathrm{Re}\langle u,v\rangle + i\cdot 4\,\mathrm{Im}\langle u,v\rangle = 4\langle u,v\rangle$.

□

Example: checking real polarization. With $u = (1,2)$, $v = (3,-1)$: $\|u+v\|^2 = \|(4,1)\|^2 = 17$, $\|u-v\|^2 = \|(-2,3)\|^2 = 13$, and $(17-13)/4 = 1 = u\cdot v$.

Example: the maximum norm. The **maximum norm** $\|(z_1,\dots,z_n)\|_\infty = \max_j |z_j|$ satisfies (i)–(iii) of Def. 44. For $u = e_1$, $v = e_2$ in $\mathbb{R}^2$: $\|u \pm v\|_\infty = \|(1,\pm 1)\|_\infty = 1$, so the left side of the parallelogram law is 2, while the right side is $2+2 = 4$.



The unit sphere of the maximum norm on $\mathbb{R}^2$ is the square. All four vectors $u, v, u+v, u-v$ lie on it, so all have norm 1: the diagonals are as long as the sides, and the parallelogram law fails ($1+1 \neq 2+2$).

The question answered. Exactly the norms satisfying the parallelogram law (Prop. 83; one direction cited), and the inner product is recovered by polarization.

Shortcuts

- **Recover an inner product from its norm**: polarization (Prop. 83). Instance: I5.
- **Test whether a norm can come from an inner product**: try the parallelogram law on $u = e_1$, $v = e_2$. Reason: Prop. 83. Instance: the max norm gives $2 \neq 4$.

Traps

- Over $\mathbb{C}$ the real polarization formula gives only $\mathrm{Re}\langle u, v\rangle$: in $\mathbb{C}$, $u = 1$, $v = i$ give $\frac{1}{4}(|1+i|^2 - |1-i|^2) = 0$, but $\langle 1, i\rangle = -i$.
- The parallelogram law must hold for all pairs: the max norm satisfies it for $u = v$ (both sides are $4\|u\|^2$), and still fails for e_1, e_2.

§2.7 Angles (Ch. 17)

The question. In $\mathbb{R}^2$ the vectors $(1,0)$ and $(1,1)$ make the angle $\pi/4$. How can an angle be defined in $\mathbb{R}^n$ or in a space of functions, where there is no picture to measure?

Definition (Def. 45 / LADR 6A.16)**.** In a *real* inner product space the **angle** $\theta \in [0, \pi]$ between nonzero u, v is defined by $\cos\theta = \langle u, v\rangle / (\|u\|\,\|v\|)$.

Cauchy–Schwarz is exactly the statement that the right side lies in $[-1, 1]$, so θ exists.

Example 68. For $(1,0)$ and $(1,1)$ in $\mathbb{R}^2$, $\cos\theta = 1/(1 \cdot \sqrt{2})$, so $\theta = \pi/4$.

The question answered. By the formula of Def. 45, which Cauchy–Schwarz makes meaningful in every real inner product space. In a complex space $\langle u, v\rangle$ need not be real, so this formula does not define an angle without an extra convention.

Traps

- The angle is defined only in a real space and for nonzero vectors: in $\mathbb{C}$, $\langle 1, i\rangle = -i$ is not a cosine.

§2.8 Optional application: attention in language models (Ch. 17)

The question. A memory stores France $\mapsto$ Paris, Germany $\mapsto$ Berlin, Italy $\mapsto$ Rome as vectors. Given a vector that asks for France, how can inner products produce mostly Paris, in a way that changes smoothly as the question vector changes?

The problem: retrieval by similarity. A memory holds m items. Item j has a **key** $k_j \in \mathbb{R}^{d_k}$, used to decide whether the item is relevant, and a **value** $\nu_j \in \mathbb{R}^{d_v}$, the information it supplies if it is used. The key and the value have different jobs, so they may have different dimensions. A request is a **query** $q \in \mathbb{R}^{d_k}$, in the same space as the keys because it is compared with them. The comparison is the inner product: a large $\langle q, k_j \rangle$ means item j matches.

What the score measures. For real vectors $\langle q, k \rangle = \|q\| \, \|k\| \cos\theta$ (Def. 45). So the score is large when k points in nearly the same direction as q, and it also grows with the lengths of q and k. It is not the angle alone; normalizing both vectors to length 1 would turn it into $\cos\theta$, but attention uses the inner product itself.

From scores to weights: softmax. We want to combine the values with weights that are positive, add up to 1, and are larger for larger scores. With scores $s_1, \ldots, s_m$, the **softmax** weights are

$$a_j = \frac{e^{s_j}}{e^{s_1} + \cdots + e^{s_m}}.$$

- Each $a_j > 0$, because $e^s > 0$; and $a_1 + \cdots + a_m = 1$, because the numerators add up to the denominator.
- The ratio of two weights depends only on the difference of scores: $a_j / a_l = e^{s_j - s_l}$. So larger scores get larger weights, and adding the same constant to every score changes nothing.
- Multiplying all scores by a large factor makes the largest score dominate: with scores 10 and 0, the weights are $e^{10}/(e^{10}+1) \approx 0.99995$ and 0.00005. So softmax is a smooth version of "pick the item with the largest score". That is the reason for the name.

The output is the weighted average of the values, $y = a_1\nu_1 + \cdots + a_m\nu_m$. Several items can contribute, in proportion to how well they match. Scaled dot-product attention uses the scores $s_j = \langle q, k_j \rangle / \sqrt{d_k}$; the factor $\sqrt{d_k}$ is explained below.

Example 69: a toy memory. Three records France $\mapsto$ Paris, Germany $\mapsto$ Berlin, Italy $\mapsto$ Rome. Choose the values $\nu_1 = e_{\text{Paris}}$, $\nu_2 = e_{\text{Berlin}}$, $\nu_3 = e_{\text{Rome}}$ (the standard basis of $\mathbb{R}^3$), the keys $k_1 = (2,0)$, $k_2 = (0,1)$, $k_3 = (-1,0)$ in $\mathbb{R}^2$, and the query $q = (1,0)$ for France. These coordinates are chosen by hand so that every step can be seen; they are not how a real model encodes countries.

1. Inner products: $\langle q, k_1 \rangle = 2$, $\langle q, k_2 \rangle = 0$, $\langle q, k_3 \rangle = -1$.
2. Here $d_k = 2$, so the scores are $2/\sqrt{2} = \sqrt{2}$, 0, $-1/\sqrt{2}$.
3. Exponentials: $e^{\sqrt{2}} \approx 4.113$, $e^0 = 1$, $e^{-1/\sqrt{2}} \approx 0.493$, with sum ≈ 5.606. Dividing gives the weights 0.734, 0.178, 0.088.
4. Output: $y \approx 0.734\, e_{\text{Paris}} + 0.178\, e_{\text{Berlin}} + 0.088\, e_{\text{Rome}}$. The best match dominates, but the others still contribute.



Left: the query and the three keys. k_1 points the same way as q and is long, so its score is largest; k_2 is perpendicular to q (score 0); k_3 points the opposite way (negative score). Right: after softmax every weight is positive and they add up to 1.

Why divide by $\sqrt{d_k}$. As the dimension d_k grows, inner products get larger, and softmax of large scores is nearly "all weight on one item". A heuristic shows how fast they grow. Suppose the coordinates q_r and k_r behave like independent random numbers with mean 0 and variance 1 (probability facts, cited).

1. Each product $q_r k_r$ has mean $E[q_r]E[k_r] = 0$ and variance $E[q_r^2 k_r^2] = E[q_r^2]E[k_r^2] = 1$, using independence.
2. The d_k products are independent, and variances of independent terms add, so $\langle q, k \rangle = \sum_{r=1}^{d_k} q_r k_r$ has variance d_k: a typical score has size about $\sqrt{d_k}$.
3. Dividing by $\sqrt{d_k}$ gives variance 1, so typical scores stay of size about 1 in every dimension, and softmax does not put almost all the weight on a single item.

Learned queries, keys and values. In a transformer, position j of the text has a current internal representation $x_j \in \mathbb{R}^{d_{\text{model}}}$. One attention head learns three linear maps $W_Q, W_K \colon \mathbb{R}^{d_{\text{model}}} \to \mathbb{R}^{d_k}$ and $W_V \colon \mathbb{R}^{d_{\text{model}}} \to \mathbb{R}^{d_v}$, and sets $q_i = W_Q x_i$, $k_j = W_K x_j$, $\nu_j = W_V x_j$. So each position offers a key (how it should be matched) and a value (what it contributes), and asks with a query. The query and key dimensions must be equal so that $\langle q_i, k_j \rangle$ is defined; nothing requires $d_v = d_k$. The keys decide which positions receive large weights; the values are the vectors that are averaged with those weights.

Self-attention. Position i compares its query with every key:

$$a_{ij} = \frac{\exp\big(\langle q_i, k_j \rangle / \sqrt{d_k}\big)}{\sum_{\ell=1}^{m} \exp\big(\langle q_i, k_\ell \rangle / \sqrt{d_k}\big)}, \qquad y_i = \sum_{j=1}^{m} a_{ij} \nu_j.$$

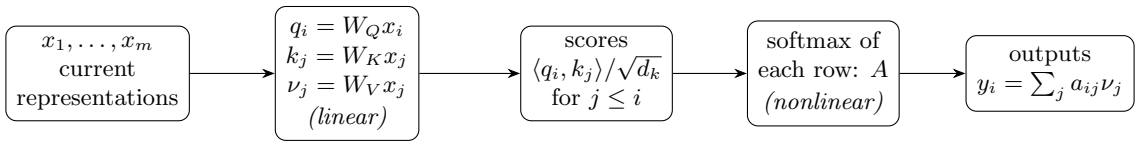
- For each i, $a_{ij} > 0$ and $\sum_j a_{ij} = 1$, so the matrix $A = (a_{ij})$ is *row*-stochastic: row i says how position i spreads its attention. (The Markov chains of Weeks 6–7 used column-stochastic matrices.)
- *Masking.* A model that writes text left to right must not let position i use later positions. So the terms with $j > i$ are removed before softmax: row i has nonzero entries only in columns $1, \dots, i$.
- *Matrix form.* Put the vectors as columns: $Q = (q_1 \cdots q_m)$, $K = (k_1 \cdots k_m)$, $V_{\text{val}} = (\nu_1 \cdots \nu_m)$. Then $S = Q^{\mathsf{T}} K / \sqrt{d_k}$ has entries $S_{ij} = \langle q_i, k_j \rangle / \sqrt{d_k}$, and A is softmax applied to each row of S. Column i of $Y = V_{\text{val}} A^{\mathsf{T}}$ is V_{val} times column i of A^{T}, which is row i of A; so it is $\sum_j a_{ij} \nu_j = y_i$. Once the weights are known, the retrieval is ordinary matrix multiplication.
- *Not linear.* A is computed from the inputs, and softmax is not linear. For instance, doubling every x_j doubles every q_i and k_j, so every score is multiplied by 4 and the weights change. Self-attention as a whole is therefore not a linear map, although each step uses linear maps.

A transformer runs several such heads in parallel, each with its own W_Q, W_K, W_V, and combines their outputs.

Example: two positions with masking. Take $W_Q = W_K = W_V = I$ on $\mathbb{R}^2$, $x_1 = (1, 0)$, $x_2 = (1, 1)$, so $d_k = 2$. Row 1 may use only position 1, so $a_{11} = 1$ and $y_1 = x_1$. Row 2 has scores $\langle x_2, x_1 \rangle / \sqrt{2} = 1/\sqrt{2}$ and $\langle x_2, x_2 \rangle / \sqrt{2} = 2/\sqrt{2}$, with weights

$$a_{21} = \frac{e^{1/\sqrt{2}}}{e^{1/\sqrt{2}} + e^{2/\sqrt{2}}} \approx 0.330, \qquad a_{22} \approx 0.670, \qquad y_2 \approx 0.330\, x_1 + 0.670\, x_2 = (1,\ 0.670).$$

If x_1, x_2 are doubled, the row-2 scores become $4/\sqrt{2}$ and $8/\sqrt{2}$, the weights become 0.056 and 0.944, and $y_2 \approx (2, 1.89)$, which is not twice $(1, 0.670)$.



One attention head. Learned linear maps produce queries, keys and values; inner products of queries with keys decide the weights; the weights mix the values. Only the softmax step is nonlinear.

The question answered. Compare the question with each key by an inner product and turn the scores into positive weights adding to 1 with softmax; the output is the weighted average of the values (Ex. 69). The weights depend smoothly on the query.

Shortcuts

- **Softmax ignores a common shift**: $a_j / a_l = e^{s_j - s_l}$. Instance: scores $(3, 1)$ and $(10, 8)$ give the same weights 0.881, 0.119.

Traps

- Attention is not linear in its inputs: doubling x_1, x_2 in the two-position example changes the weights from $0.330, 0.670$ to $0.056, 0.944$.
- The attention matrix is row-stochastic, not column-stochastic as in Weeks 6–7.

§3 Problem set

HW9 P1. Let P be a projection on a finite-dimensional vector space, so that $P^2 = P$. Prove that $\operatorname{tr} P = \operatorname{rank} P$. You may choose a basis adapted to the decomposition $V = \operatorname{null} P \oplus \operatorname{range} P$.

Hint: $V = \operatorname{null} P \oplus \operatorname{range} P$ (T5); in a basis of $\operatorname{null} P$ followed by a basis of $\operatorname{range} P$ the matrix of P is diagonal with entries 0 and 1.

HW9 P2. Let $T \in \mathcal{L}(\mathbb{C}^5)$ have minimal polynomial $m_T(x) = (x-2)^2(x+1)$.

(a) Determine all possible algebraic multiplicities of the eigenvalues 2 and -1.

(b) Determine all possible values of $\operatorname{tr} T$.

(c) For each possible trace, exhibit a Jordan normal form realizing it.

Hint: the exponent of $x - \lambda$ in m_T is the size of the largest Jordan block for λ (T6); the block sizes add up to 5; then Prop. 79.

HW9 P3.

(a) Prove that every nilpotent operator on a finite-dimensional complex vector space has trace 0.

(b) Give a nonzero operator with trace 0 that is not nilpotent.

(c) Give two 2×2 matrices A, B for which $AB - BA \neq 0$, and verify directly that $\operatorname{tr}(AB - BA) = 0$.

Hint: (a) the only eigenvalue of a nilpotent operator is 0, then Prop. 79; (b) a diagonal matrix; (c) the matrix units E_{12}, E_{21}.

HW9 P4. For each of the following formulas on $\mathbb{R}^2$, decide whether it defines an inner product. Justify your answer.

$$\langle x, y\rangle_1 = x_1y_1 + 2x_2y_2, \qquad \langle x, y\rangle_2 = x_1y_1 - x_2y_2, \qquad \langle x, y\rangle_3 = (x_1 + x_2)(y_1 + y_2).$$

Hint: all three are bilinear and symmetric, so test positivity; use the 2×2 test in the Shortcuts of Section 2.4, or find a nonzero x with $\langle x, x\rangle \leq 0$.

HW9 P5. On $\mathcal{P}_2(\mathbb{R})$ define $\langle p, q\rangle = \int_{-1}^{1} p(x)q(x)\,dx$.

(a) Prove that this is an inner product.

(b) Compute the norms of 1, x, and x^2.

(c) Determine which pairs among $1, x, x^2$ are orthogonal.

Hint: positivity: p^2 is continuous and ≥ 0, so (T7), then a polynomial with infinitely many roots is 0; for (c), odd functions have integral 0 on $[-1, 1]$.

HW9 P6. Use the Cauchy–Schwarz inequality to prove that for all real numbers a, b, c, $(a+b+c)^2 \leq 3(a^2+b^2+c^2)$. Determine exactly when equality holds.

Hint: Cauchy–Schwarz (Prop. 82) for (a, b, c) and $(1, 1, 1)$; the equality case of Prop. 82.

HW9 P7. (optional) Prove the parallelogram identity $\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$ in every real or complex inner-product space.

Hint: expand $\langle u \pm v, u \pm v\rangle$ as in the proof of Prop. 83.

I1 (the trace inner product). For real $n \times n$ matrices define $\langle A, B\rangle = \operatorname{tr}(A^{\mathsf{T}}B)$.

(a) Show that $\langle A, A\rangle = \sum_{i,j} a_{ij}^2$, and deduce that $A^{\mathsf{T}}A = 0$ only if $A = 0$.

(b) Show that $\langle \cdot, \cdot \rangle$ is an inner product on $\mathbb{R}^{n\times n}$.

(c) Use Cauchy–Schwarz to show $\big(\operatorname{tr}(A^{\mathsf{T}}B)\big)^2 \leq \operatorname{tr}(A^{\mathsf{T}}A)\operatorname{tr}(B^{\mathsf{T}}B)$.

Hint: compute the diagonal entries of $A^{\mathsf{T}}B$ (as in Prop. 78); this is the dot product on $\mathbb{R}^{n^2}$.

I2 (trace of a real matrix). A real 3×3 matrix A has eigenvalues 2 and $1+i$. Find $\operatorname{tr} A$. If moreover A is invertible, can $\operatorname{tr} A^{-1}$ be found from this information?

Hint: non-real eigenvalues of a real matrix come in conjugate pairs (Section 2.2, "Real matrices"); three distinct eigenvalues give an eigenbasis.

I3 (commutators). (a) Find 2×2 matrices A, B with $AB - BA = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. (b) Show there are no 2×2 matrices with $AB - BA = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$.

Hint: (a) the matrix units E_{12}, E_{21}; (b) Prop. 80.

I4 (Cauchy–Schwarz). (a) For real $x_1, \dots, x_n$, prove that the square of the average is at most the average of the squares. (b) For continuous $f\colon [0,1] \to \mathbb{R}$, prove

$$\left(\int_0^1 x f(x)\,dx\right)^2 \le \frac{1}{3}\int_0^1 f(x)^2\,dx,$$

and find all f for which equality holds.

Hint: Cauchy–Schwarz with $(1,\dots,1)$, and with the L^2 product and $g(x) = x$; equality case of Prop. 82.

I5 (polarization). The function $\|(x,y)\| = \sqrt{x^2 + xy + y^2}$ is a norm on $\mathbb{R}^2$ that satisfies the parallelogram law. Use polarization to find the inner product it comes from, and check that the result is an inner product.

Hint: real polarization identity (Prop. 83), then the 2×2 positivity test (Shortcuts of Section 2.4).

I6 (angles in high dimensions). In $\mathbb{R}^n$ with the dot product, find the angle between $(1,1,\dots,1)$ and e_1. What happens as $n \to \infty$?

Hint: Def. 45; $\|(1,\dots,1)\| = \sqrt{n}$.

I7 (attention). In the toy memory of Ex. 69, (a) compute the weights for the query $q = (0,1)$; (b) show that replacing q by tq with $t \to \infty$ makes the weights for the query $(1,0)$ tend to $(1,0,0)$; (c) show that adding the same vector w to all three keys does not change the weights, but multiplying all three keys by 2 does.

Hint: weights depend only on differences of scores ($a_j/a_l = e^{s_j - s_l}$).